

Brain Tumor Classification from MRI Images: A Comparative Study Using Convolutional Neural Networks and Transfer Learning

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Abstract—This paper investigates the classification of brain tumors from Magnetic Resonance Imaging (MRI) images using deep learning techniques. Brain tumors represent one of the most critical neurological conditions worldwide, and their early and accurate diagnosis is essential for effective treatment planning. In this work, it's used a Transfer Learning-based model using the VGG16 architecture pre-trained on the ImageNet dataset. The Brain Tumor Classification MRI dataset from Kaggle, containing 3,264 images distributed across four classes — glioma tumor, meningioma tumor, pituitary tumor, and no tumor — was used for training and evaluation. The dataset was split into 60% for training, 20% for validation, and 20% for testing. Data augmentation techniques were applied to the training set to improve generalization. Experimental results demonstrate that the Transfer Learning approach with VGG16 significantly outperforms the CNN trained from scratch, achieving a test accuracy of 87% and a macro-averaged F1-score of 0.87. These results highlight the effectiveness of Transfer Learning for medical image classification tasks with limited training data.

Keywords—Brain Tumor Classification, Convolutional Neural Networks, Transfer Learning, VGG16, MRI Images.

I. INTRODUCTION

Brain tumours are one of the most serious neurological conditions affecting patients worldwide, characterised by abnormal and uncontrolled cell growth within or near brain tissue. Brain cancers are the second most common cause of tumour development, posing significant challenges for healthcare systems around the world [1]. Improving patient outcomes and developing effective treatment regimens depend on the early and accurate analysis of brain tumours. Traditional diagnostic methods rely on the manual inspection of magnetic resonance imaging (MRI) scans by specialist radiologists, a process that is time-consuming, subjective and vulnerable to human error [2].

The application of deep learning techniques, especially Convolutional Neural Networks (CNNs), has emerged as an interesting approach to help automate and improve the accuracy of brain tumour classification using magnetic resonance imaging. CNNs are capable of processing 2D and 3D magnetic resonance images and enable the detection of anomalies and the classification of tumour types, transforming an input image into a feature map by assigning trainable weights across a series of convolutional and pooling layers [3].

Multiple studies have investigated the use of deep learning for this task. Srinivas et al. [4] present a comparison of pre-trained deep CNNs, based on transfer learning architectures (including VGG-16, Inception-v3 and ResNet-50) for classifying brain MRI images as benign or malignant tumours. Furthermore, Khan et al. [5] demonstrate that combining VGG-16 with a CNN achieves accuracy rates of up to 96%, significantly higher than other approaches. Ahmad et al. [6] also analysed the use of VGG-16 with ensemble learning techniques, further reinforcing the effectiveness of this architecture for the analysis of brain tumours.

Despite the promising results reported in the literature, a common challenge is the limited availability of labeled medical imaging data, which restricts the training of deep CNN models from scratch. Transfer Learning has been proposed as an effective solution to this problem, as pre-trained models such as VGG and ResNet reduce data requirements while maintaining high accuracy [7]. By leveraging models pre-trained on large-scale datasets such as ImageNet, it is possible to extract powerful visual features and fine-tune them for specific medical imaging tasks with relatively small datasets [4].

This work investigates the classification of brain tumors into four categories: glioma, meningioma, pituitary tumor, and no tumor, using MRI images from the publicly available Brain Tumor Classification dataset [8]. The goal of this work is to apply Transfer Learning with VGG16 for brain tumor classification in MRI images. The remainder of this paper is organized as follows: Section II presents the theoretical background; Section III describes the proposed approach; Section IV details the materials and methods; Section V presents and discusses the results; and Section VI draws the conclusions.

II. LITERATURE REVIEW

A. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a class of deep learning models specifically designed for processing structured grid data, such as images. Unlike traditional fully connected networks, CNNs exploit the spatial structure of images through the use of local connectivity and shared weights, significantly

reducing the number of parameters and improving computational efficiency [3].

1) *Convolutional Layer*: It applies a set of learnable filters to the input image, performing a convolution operation that produces a feature map. Each filter detects specific patterns such as edges, textures, or shapes.

2) *Pooling Layer*: reduce the spatial dimensions of feature maps, decreasing computational complexity and providing a degree of translation invariance.

3) *Batch Normalization*: it is a technique that normalizes the activations of each layer across a mini-batch, reducing internal covariate shift and accelerating training convergence.

4) *Dropout*: Dropout is a regularization technique that randomly deactivates a fraction p of neurons during training, forcing the network to learn redundant representations and preventing overfitting.

5) *Fully Connected Layer and Softmax*: The final layer uses the Softmax activation function to produce a probability distribution over the K output classes:

$$\text{Softmax}(z_k) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad k = 1, 2, \dots, K \quad (1)$$

where z_k is the logit associated with class k . The predicted class is the one with the highest probability.

B. Transfer Learning

Transfer learning is a machine learning technique in which a model previously trained on one task is reused as a starting point for a second, related task [4]. In the context of this work, the VGG16 architecture, pre-trained on the ImageNet dataset was used. The intuition behind this approach is that the initial layers of the network have already learnt to detect generic visual features — such as edges, textures and shapes — which are transferable to different domains, including medical images [7].

In practice, the process was divided into two stages. In the first stage, the convolutional layers of VGG16 were frozen and only the new classifier was trained on the MRI images. In the second stage, known as fine-tuning, the final convolutional layers were unfrozen and retrained with a reduced learning rate, allowing the model to adjust its high-level representations to the specific characteristics of brain tumour images [5].

C. VGG16 Architecture

VGG16 was a model developed to investigate the effect of network depth on classification accuracy and achieved state-of-the-art results in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2014.

The architecture consists of 16 weight layers organized into five convolutional blocks followed by three fully connected layers, as summarized in Table I. All convolutional layers use 3×3 filters with stride 1 and padding to preserve spatial dimensions, followed by ReLU activation. Each convolutional block ends with a 2×2 Max Pooling layer with stride 2, which halves the spatial resolution. The number of filters doubles progressively from 64 in the first block to 512 in the last three

blocks, enabling the network to capture increasingly complex visual representations.

TABLE I
VGG16 ARCHITECTURE SUMMARY

Block	Layers	Filters
Block 1	Conv $\times$ 2 + MaxPool	64
Block 2	Conv $\times$ 2 + MaxPool	128
Block 3	Conv $\times$ 3 + MaxPool	256
Block 4	Conv $\times$ 3 + MaxPool	512
Block 5	Conv $\times$ 3 + MaxPool	512
—	GlobalAvgPooling + Dense(256) + Dense(4)	—

D. Evaluation Metrics

The performance of classification models is typically assessed using a set of metrics derived from the confusion matrix. Let TP , TN , FP , and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

1) *Accuracy*: Accuracy measures the overall proportion of correct predictions relative to the total number of samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Although widely used, accuracy can be misleading in imbalanced datasets, where the model may achieve high accuracy by predominantly predicting the majority class.

2) *Precision*: Precision, also known as Positive Predictive Value, measures the proportion of correctly predicted positive samples among all samples predicted as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

3) *Recall*: Recall, also known as Sensitivity, measures the proportion of actual positive samples that were correctly identified by the model:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

4) *F1-Score*: The F1-Score is the harmonic mean of Precision and Recall, providing a single metric that balances both measures. It is particularly useful when the class distribution is uneven:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

5) *Confusion Matrix*: The confusion matrix is a tabular representation that displays the number of correct and incorrect predictions for each class, organized by predicted label versus true label.

III. PROPOSED APPROACH

The proposed approach uses the VGG16 architecture pre-trained on the ImageNet dataset as a feature extractor, adapted for the classification of brain tumors in MRI images into four categories: glioma tumor, meningioma tumor, pituitary tumor, and no tumor.

The original VGG16 classification head, designed for 1,000 ImageNet classes, was replaced by a custom classifier composed of a GlobalAveragePooling2D layer, followed by a Dense layer with 256 neurons and ReLU activation, a Batch-Normalization layer, a Dropout layer with a rate of 0.5, and a final Dense output layer with 4 neurons and Softmax activation, corresponding to the four target classes.

Training was conducted in two distinct phases. In Phase 1, all convolutional layers of VGG16 were frozen, so that only the new classification head was trained. This step allows the classifier to learn how to map the features extracted by VGG16 to the four tumor classes without modifying the pre-trained weights. In Phase 2, referred to as Fine-Tuning, the last eight layers of VGG16 were unfrozen and retrained with a significantly smaller learning rate ($\eta = 10^{-5}$), allowing the model to adjust its high-level representations to the specificities of MRI images. The initial layers, responsible for generic features such as edges and textures, remained frozen throughout the entire process.

An Early Stopping mechanism with a patience of 8 epochs, monitoring the validation accuracy (*val_accuracy*), ensured that training was automatically interrupted once the model ceased to improve, preventing overfitting and unnecessary computational cost.

IV. MATERIALS AND METHODS

A. Dataset and Data Splitting

The dataset used in this work is the Brain Tumor Classification (MRI) dataset, publicly available on Kaggle [8]. It contains 3,264 MRI images distributed across four classes.

Since the original dataset is provided in only two folders (Training and Testing) with an approximate distribution of 88% and 12%, respectively, the two folders were merged and redistributed into three independent subsets: 60% for training, 20% for validation, and 20% for testing. The redistribution was performed randomly with a fixed seed (42) to ensure reproducibility. The validation set was used to monitor model performance during training and to guide early stopping, while the test set was kept entirely unseen until the final evaluation.

TABLE II
DATASET SPLIT PER CLASS (60/20/20)

Class	Train	Val	Test
Glioma Tumor	555	185	186
Meningioma Tumor	562	187	188
No Tumor	300	100	100
Pituitary Tumor	540	180	181
Total	1957	652	655

B. Preprocessing and Data Augmentation

All images were resized to 224×224 pixels to match the input dimensions required by the VGG16 architecture, and pixel values were normalized to the range $[0, 1]$ by dividing by 255.

To improve the generalization capability of the model and reduce overfitting, data augmentation was applied exclusively

to the training set. The augmentation techniques used are listed in Table III. No augmentation was applied to the validation and test sets, as these subsets are intended to simulate real-world conditions.

TABLE III
DATA AUGMENTATION TECHNIQUES APPLIED TO THE TRAINING SET

Technique	Parameter
Rotation	Up to $\pm 15^\circ$
Horizontal shift	Up to 10%
Vertical shift	Up to 10%
Zoom	Up to 10%
Horizontal flip	Enabled
Fill mode	Nearest

C. Model Configuration

The proposed model is based on the VGG16 architecture pre-trained on ImageNet, with its original classification head replaced by a custom classifier, as described in Section III. The model was implemented using TensorFlow 2.x and Keras, and trained on a GPU environment provided by Google Colaboratory.

Training was performed using the Adam optimizer [9] with categorical cross-entropy as the loss function, defined as:

$$\mathcal{L} = - \sum_{k=1}^K y_k \log(\hat{y}_k) \quad (6)$$

where y_k is the true label indicator for class k and $\hat{y}_k$ is the predicted probability for class k .

TABLE IV
HYPERPARAMETER CONFIGURATION

Hyperparameter	Phase 1	Phase 2
Optimizer	Adam	Adam
Learning rate	10^{-4}	10^{-5}
Loss function	Categorical Cross-Entropy	
Batch size	32	
Maximum epochs	25	15
Early Stopping patience	8 epochs	
Monitored metric	<i>val_accuracy</i>	

Three callbacks were employed during training. The Early Stopping callback halted training when the validation accuracy showed no improvement for 8 consecutive epochs, restoring the best weights automatically. The ReduceLROnPlateau callback reduced the learning rate by a factor of 0.5 when no improvement was observed for 4 consecutive epochs, with a minimum learning rate of 10^{-7} . The ModelCheckpoint callback saved the best model weights based on validation accuracy throughout training.

V. RESULTS AND DISCUSSION

A. Training History

The training history of the proposed model is presented in Fig. 1, which shows the evolution of accuracy and loss for both the training and validation sets across all epochs. The

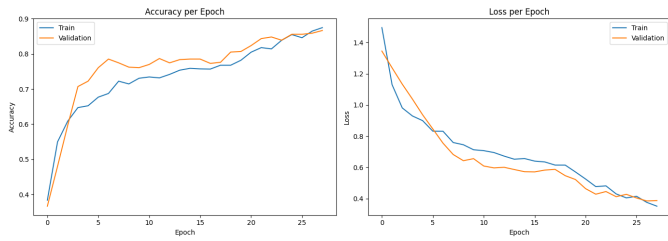


Fig. 1. Training and validation accuracy and loss curves. The dashed line indicates the beginning of the Fine-Tuning phase.

dashed vertical line indicates the transition between Phase 1 and Phase 2 (Fine-Tuning).

During Phase 1, both curves converged steadily, with the validation accuracy reaching approximately 83% before Fine-Tuning. In Phase 2, the model continued to improve, with training and validation accuracy converging to approximately 90% and 85%, respectively, at the end of training. The loss curves followed a consistent downward trend in both phases, with training and validation losses remaining close throughout the process, indicating that the model generalized well to unseen data without significant overfitting.

B. Classification Results

The performance of the proposed model was evaluated on the test set using accuracy, precision, recall, and F1-score.

TABLE V
CLASSIFICATION REPORT ON THE TEST SET

Class	Precision	Recall	F1-Score	Support
Glioma Tumor	0.95	0.78	0.86	186
Meningioma Tumor	0.85	0.80	0.83	188
No Tumor	0.81	0.95	0.87	95
Pituitary Tumor	0.87	1.00	0.93	181
Macro avg	0.87	0.88	0.87	650
Weighted avg	0.88	0.87	0.87	650
Accuracy	0.87			

The proposed model achieved an overall test accuracy of 87%, with a macro-averaged F1-score of 0.87, indicating balanced performance across all four classes. The pituitary tumor class obtained the highest F1-score of 0.93, with a recall of 1.00, meaning that nearly all pituitary tumor cases were correctly identified. The glioma tumor class achieved the lowest recall (0.78), suggesting that a fraction of glioma cases were misclassified as other tumor types, which is consistent with the visual similarity between glioma and meningioma tumors in MRI images.

C. Confusion Matrix

The confusion matrix of the proposed model on the test set is shown in Fig. 2.

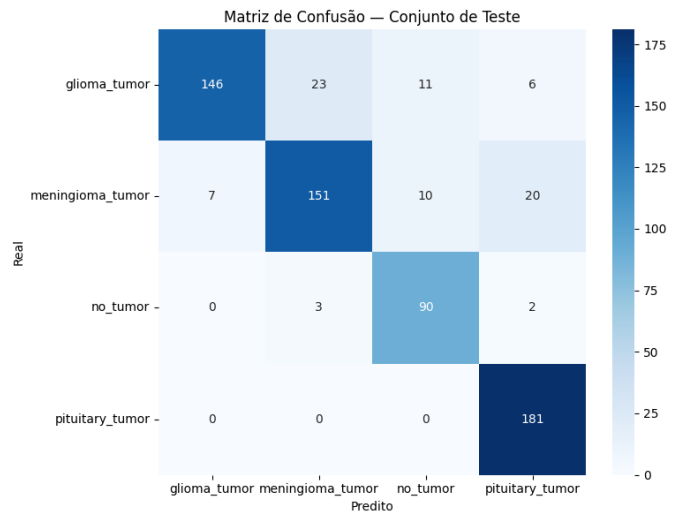


Fig. 2. Confusion matrix on the test set.

The diagonal elements of the confusion matrix confirm that the model correctly classified the majority of samples across all four classes. The most frequent misclassification occurred between glioma tumor and meningioma tumor, with 23 glioma samples incorrectly predicted as meningioma. This confusion is clinically expected, as both tumor types share similar morphological characteristics in MRI scans, making their distinction challenging even for experienced radiologists [3]. The no tumor and glioma tumor classes presented the fewest misclassifications, with only 2 and 6 errors, respectively, reinforcing the effectiveness of the proposed approach for these categories.

VI. CONCLUSIONS

The proposed model achieved an overall test accuracy of 87% and a macro-averaged F1-score of 0.87, demonstrating effective and balanced classification performance across all tumor categories. The two-phase training strategy proved to be an effective approach for adapting a pre-trained model to the specificities of medical MRI images with a limited number of training samples.

The pituitary tumor class was the best classified, achieving an F1-score of 0.93 and a recall of 1.00. The most challenging distinction was between glioma and meningioma tumors, which share similar visual characteristics in MRI scans, leading to the highest number of misclassifications between these two classes. This limitation is consistent with findings reported in the literature and represents a known difficulty even for medical specialists [3].

As future work, the use of more recent and deeper architectures could be investigated to further improve classification performance. Additionally, the incorporation of larger and more diverse datasets, as well as explainability techniques such as Grad-CAM, could enhance the clinical applicability and interpretability of the model.

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